

JEREMIAH NYAGAH NATIONAL POLYTECHNIC.
DEPARTMENT: ELECTRICAL AND ELECTRONIC ENGINEERING.
CLASS: DCBET/EE/24M/6
CAT 1 AND 2 COMBINED
SUBJECT: POWER GENERATION

SECTION A (40 MARKS)

*Answer **all** questions in this section.*

1. Define the following terms as used in power generation. (4 Marks)
 - a. Power station
 - b. Alternator
 - c. Transformer
 - d. Water head
2. Differentiate light energy from sound energy. (2 Marks)
3. Differentiate chemical energy from nuclear energy. (2 Marks)
4. State three forms of kinetic energy. (3 Marks)
5. State two forms of potential energy. (2 Marks)
6. How much energy in joules is in 10 units of electricity? (2 Marks)
7. Electrical energy is said to be the most superior form of energy. Give five reasons why. (5 Marks)
8. Energy exists in free form in nature, state five sources of energy. (5 Marks)
9. A modern power station has electrical equipment, state three items comprising electrical equipment. (3 Marks)
10. A steam power station has an overall efficiency of 20% and 0.6 kg of coal is burnt per KWh of electrical energy generated. Calculate the calorific value of fuel. (4 Marks)
11. Power stations must be located strategically for maximum efficiency, State four factors to consider when selecting the site for a steam power station. (4 Marks)
12. Hydro-electric power stations provide most of electricity in Kenya, State four advantages of hydro-electric power stations. (4 Marks)

SECTION B (60 MARKS)

*Answer any **THREE** questions in this section.*

11. (a) Diesel power plants are not very popular, Explain four reasons why . (8 Marks)

(b) Using a schematic arrangement diagram, explain the components that constitute a hydro-electric power station. (12 Marks)

12. (a) A nuclear power station must be located in appropriate sites, explain five factors to consider when choosing a site for such a station. (10 Marks)

(b) A hydro-electric power station has a reservoir of 2.4 square kilometers and capacity $5 \times 10^6 \text{ m}^3$. The effective head of water is 100 meters. The penstock, turbine and generation efficiencies are respectively 95%, 90% and 85%.

i. Calculate the total electrical energy that can be generated from the power station. (5 Marks)

ii. If a load of 15,000 KW has been supplied for 3 hours, find the fall in reservoir level. (5 Marks)

13. (a) A diesel engine power plant has one 700KW and two 500KW generating units. The fuel consumption is 0.28 kg per KWh and the calorific value of fuel oil is 10200 kcal/kg. Estimate;

i. The fuel oil required for a month of 30 days

ii. Overall efficiency (8 Marks)

(b) Using a schematic arrangement diagram, explain the operation of a gas turbine power plant. (12 Marks)

14. Give the comparison of steam power plant, hydro-electric power plant, diesel power plant and nuclear power plant on the basis of operating cost, initial cost, efficiency, maintenance cost and availability of source of power. (20 Marks)